Mohammadali Zayandehroodi

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Mohammadali Zayandehroodi

EDUCATION

Ph.D. in Transportation Engineering *Oct 2021 – Sep 2025*

Iran University of Science & Technology, Tehran, Iran

Thesis: *Ranking and Selection of Safety Countermeasures in Highway-Rail Grade Crossings*

Supervisors: Dr. Barat Mojaradi & Dr. Morteza Bagheri

M.Sc. in Transportation Engineering *Oct 2019 – Sep 2021*

Shahid Bahonar University of Kerman, Iran

Thesis: *Managing countermeasures to improve road network safety with an economic approach*

Supervisors: Dr. Navid Nadimi & Dr. Abdel-Majid Jalaee

B.Sc. in Civil Engineering *Oct 2014 – Sep 2019*

Shahid Bahonar University of Kerman, Iran

EXPERIENCE

Visiting Researcher *Feb 2025 – Aug 2025*

Institute for Transport Planning and Systems, ETH Zurich, Zurich, Switzerland

Project: Generation of Synthetic Rear-End Crash Scenarios

PI: Anastasios Kouvelas

Visiting Researcher *Apr 2022 – Jun 2022*

Larson transportation institute, Penn State University, PA, USA

Host: Dr. Faeze Ghofrani

Teaching Assistant Oct 2023 – *Sep 2024*

Iran University of Science and Technology

Transportation System Modeler *Aug 2021 – Mar 2022*

Tehran International Chamber of Commerce, Tehran, Iran

TEACHING

Teaching, Shahid Bahonar University of Kerman, Kerman, Iran *Oct 2025 – Present*

Course: Traffic Engineering

Guest Lecturer, ETH Zurich, Zurich, Switzerland *Apr 2025*

Course: Traffic Engineering

Teaching Assistant, IUST, Tehran, Iran *Oct 2024 – Feb 2025*
Course: Transportation Engineering

Teaching Assistant, IUST, Tehran, Iran *Mar 2024 – Aug 2024*
Course: Airport Engineering

PUBLICATIONS

1. **Zayandehroodi, M.**, Mojaradi, B., & Bagheri, M. (2025). Improving reliability of safety countermeasure evaluation at highway-rail grade crossings through aleatoric uncertainty modeling with machine learning techniques. *Reliability Engineering & System Safety*, 261, 111082.

2. **Zayandehroodi, M.**, Mojaradi, B., & Bagheri, M. (2025). Evaluating the effectiveness of safety countermeasures at highway–railway grade crossing based on a machine learning framework. *Traffic Injury Prevention*, 26(1), 120–127.

3. Nadimi, N., **Zayandehroodi, M**., Mansourifar, F., & Shaaban, K. (2026). A multi-stage approach for assessing factors influencing pedestrian safety. *Transportation Research Part F: Traffic Psychology and Behaviour*, 120,103672.

4. Nadimi, N., Asadamraji, M., **Zayandehroodi, M.** et al. (2026). Enhancing the attractiveness of urban bus services in medium-sized cities: a structural equation modeling approach. *Innovative Infrastructure Solutions*. 11, 274.

5. Nadimi, N., Mansourifar, F**., Zayandehroodi, M. A.**, Litman, T., & Asadamraji, M. (2026). Air Pollution and Transportation: Evaluating Different Factors from a COVID-19 Outbreak Perspective. *Journal of Urban Planning and Development*, *152*(2), 04026010.

6. Nadimi, N., Zayandehroodi, M**.**, Jalilzadeh, A., **Zayandehroodi, M** & Litman, T. (2025). Facilitating elderly tourism by sustainable transportation. *Research in Transportation Economics*, 112, 101584.

7. Nadimi, N., Loo, B. P., Mansourifar, F., **Zayandehroodi, M.**, & Kazemi, M. (2025). Earthquake-related evacuation transportation: Insights from Kerman, Iran. *Cities*, 158, 105713.

8. Nadimi, N., **Zayandehroodi, M.**, Rahmani, F., Asadamraji, M., & Litman, T. (2024). Evaluation of the impact of Covid-19 on transport sustainability in Iran. *Proceedings of the Institution of Civil Engineers-Engineering Sustainability*, 40, 1–25.

9. Nadimi, N., **Zayandehroodi, M.**, Camporeale, R., & Asadamraji, M. (2023). A framework for providing information about parking spaces. *Sustainability*, 15(19), 14505.

CONFERENCES PRESENTATION

- **Transportation Research Board (TRB) 2026**
  Washington, D.C., USA | 2026

  Zayandehroodi, M., El-Baklish, S., Makridis, K., & Kouvelas, A. (2026). *Performance Evaluation of Surrogate Safety Measures with Synthetic Crash Scenarios*.

- **25th Swiss Transport Research Conference (STRC) 2025**
  Locarno, Switzerland | 2025

  Zayandehroodi, M., Mojaradi, B., Bagheri, M., & Kouvelas, A. (2025). *Evaluating safety countermeasures at highway-railway grade crossings: A review*. Presentation at STRC 2025.

- **13th Symposium of the European Association for Research in Transportation (hEART) 2025**
  Munich, Germany | 2025

  Attended to stay current on trends in transport safety and control.

- **International Conference on Intelligent Traffic and Transportation (ICITT) 2025**
  Italy | 2025

  Zayandehroodi, M., Mojaradi, B., & Bagheri, M. *A Novel LSTM-CART Hybrid Approach for Modeling Accident Severity at Highway-Rail Grade Crossings Using Deep Learning Architecture.*

PUBLICATIONS IN PREPARATION

- Jafari, N., **Zayandehroodi, M.**, Amini, Z. *"Traffic Control for Urban Intersections: A Simulation-Based Evaluation of Traffic Management and Disruption Mitigation" (in progress for Transportation Research Part C: Emerging Technologies).*

- **Zayandehroodi, M.,** SK El-Baklish, Kouvelas, A., Makridis, M. *"Generation of Synthetic Rear-End Crash Scenarios for Evaluating Reaction-Based Surrogate Safety Measures " (in progress for Transportation Research Part C: Emerging Technologies).*

- Shan, J., **Zayandehroodi, M.** *“Using Large Language Models for Disaster Prediction in Eurasia.” (In progress for Reliability Engineering & System Safety).*

MENTORSHIP

Graduate research mentorship

Zahra Sadat Sanagostar (M.S. student, School of Railway Engineering, 2025

Iran University of Science & Technology, Tehran, Iran)

Investigating violations at Highway-Rail Grade Crossings using video analysis

Leif Keller (M.S. student, Traffic Engineering Group (SVT) 2025

Department of Civil, Environmental and Geomatic Engineering, ETH Zurich, Switzerland)

Unravelling the effect of Tempo 30 on reducing urban traffic accidents

Armina Jalilzadeh (M.S. student, Shahid Bahonar University of Kerman, Iran) 2024

Relation of evacuation in transportation

Facilitating elderly tourism by sustainable transportation

Farnaz Mostofi (M.S. student, School of Railway Engineering, 2024

Iran University of Science & Technology, Tehran, Iran)

LLM-Guided Modeling for the Economic Evaluation of Network-Wide Safety Interventions

Undergraduate research mentorship

Parmida Nemati (M.S. student, School of Industrial Engineering, 2026

Iran University of Science & Technology, Tehran, Iran)

Traffic light control at intersections, with interruption power outage interference

Ali Esmaeili (M.S. student, School of Railway Engineering, 2026

Iran University of Science & Technology, Tehran, Iran)

Conflict analysis at Highway-Rail Grade Crossings using machine learning, with uncertainty

HONORS AND AWARDS

Ranked 1st among PhD students in the Civil Engineering Department 2025

Iran University of Science and Technology

Member of Iran's National Elites Foundation 2022-2026

Visiting research grants, ETH, IVT 2025

15000CHF

REVIEWER

Journal of Traffic Injury Prevention (3) Journal of Transportation Letters (4) Journal of Reliability Engineering & System Safety (2)

SKILLS

Research Areas

Traffic engineering and traffic operations

Traffic signal control and optimization

Large language models (LLMs) for transportation

Transportation systems modeling and simulation

Software

Python, MATLAB, VISSIM, SUMO, R, ArcGIS, SPSS, etc.